

Team 2 Design Pack:

NFR, FR, UR and KPI

This is a reference list of all the identified targets, objectives and requirements we need for the project

Key Performance Indicators

- Expressions Of Interest
- Account interest
- System scalability and performance
- Accessibility

User Requirements

- Downloadable content
- Accessibility features for users with sight loss
- *Social* features

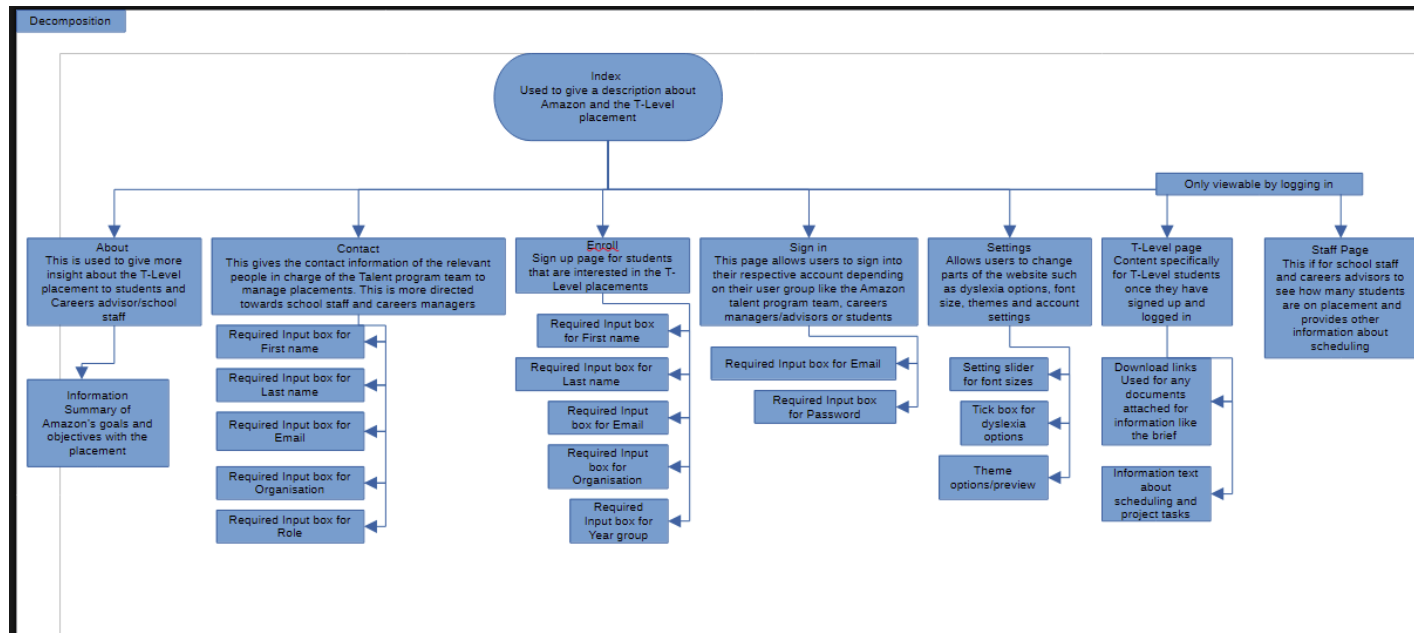
Functional Requirements

- General information and guidance on T-Level's and placements at Amazon
- Access to digital content to support customers in the form of webpages
- Expressions of interest
- *Social* features
- Accessibility
- Branding

Non-Functional Requirements

- Fast webpages
- Encryption
- Database robustness
- Data security

Overview / decomposition diagram



Colour Scheme

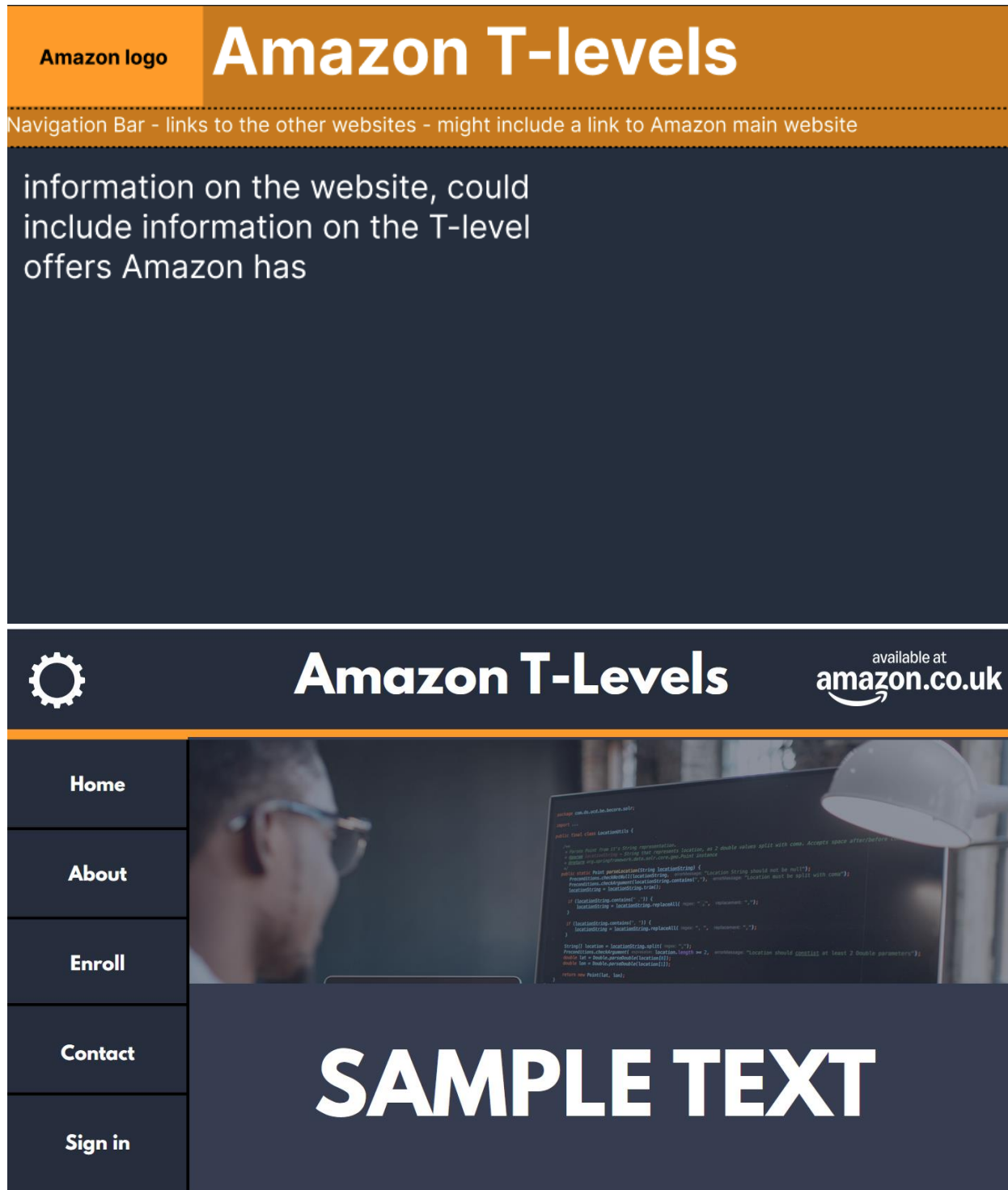


For this project, we will use the following colours which are used by amazon:

- This is their dark blue(232F3E), and their orange (FF9900). This is because this is their iconic colours and it is used on every website that is owned by Amazon.
- On top of these colours, we plan to use lighter versions of this colour to make different parts of our website standout such as the navigation bar and other items you can interact with.
- #c77700 - This is a darker orange colour; this could be used for hovering over buttons when users are going to interact with them so that it is highlighted as a

button and not just plain text. This is done intentionally to make the website feel more vibrant to create a better user experience.

Visual interface Designs





About T-Levels

available at
amazon.co.uk

Home

For T-Level Students

T-Level summary

About

Enroll

For Careers Advisors

Placement Summary

Contact

Sign in



Enroll for T-Levels

available at
amazon.co.uk

Home

FOR STUDENTS ONLY!

First Name:

About

Last Name:

School Email:

Enroll

Password

School:

Contact

Year Group:

Sign in

SUBMIT



Contact

available at
amazon.co.uk

Home

FOR CAREERS ADVISORS

First Name:

About

Last Name:

Enroll

School Email:

Contact

School:

Sign in

SUBMIT

Amazon Email:

SAMPLE TEXT@gmail.com

Enquiries Email:

SAMPLE TEXT@gmail.com

Other Emails..



Sign In

available at
amazon.co.uk

Home

Email address:

About

Password:

Enroll

Remember me: ☐

Contact

LOGIN

Sign in



Settings

available at
amazon.co.uk

Home

Font Size

About

Dyslexia-Friendly Font ☐

Enroll

High Contrast ☐

Contact

Dark Mode ☐

Light Mode ☐

Sign in

Account Settings:

Change Email:

Change Password:



T-Level Student

available at
amazon.co.uk

Home

About

Enroll

Contact

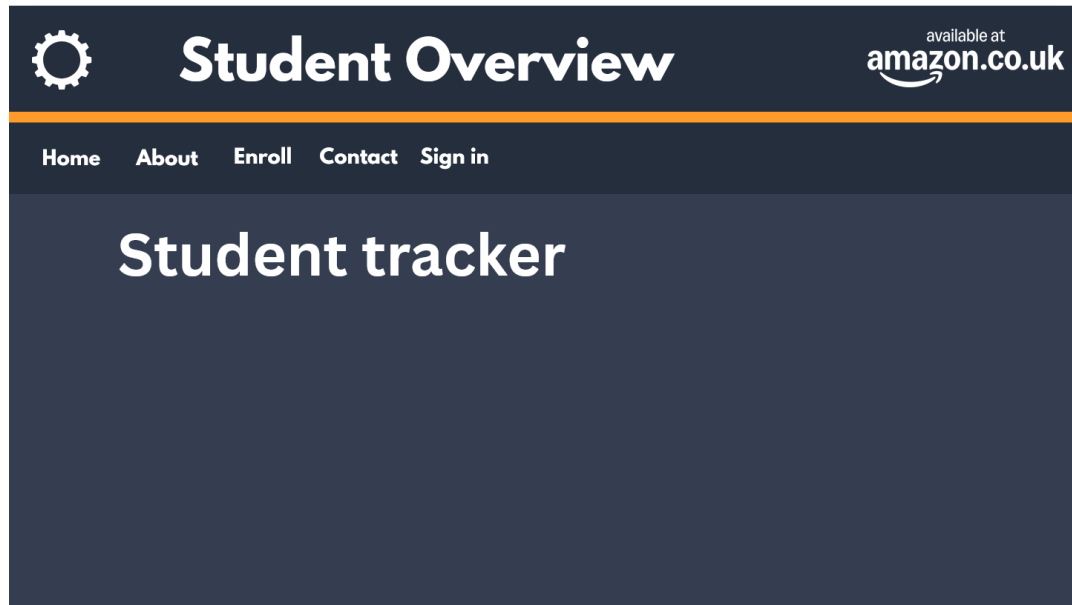
Sign in

Placement details

Long paragraph about placements

Download links

Student resources

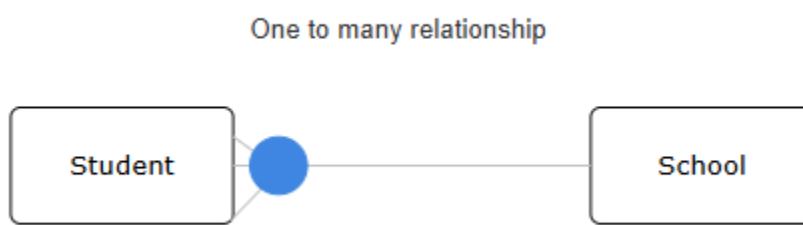


ER Diagram for Database

Factors:

- Email
- Phone number
- First name
- Surname
- School
- School email
- School phone
- Year group
- Pathway

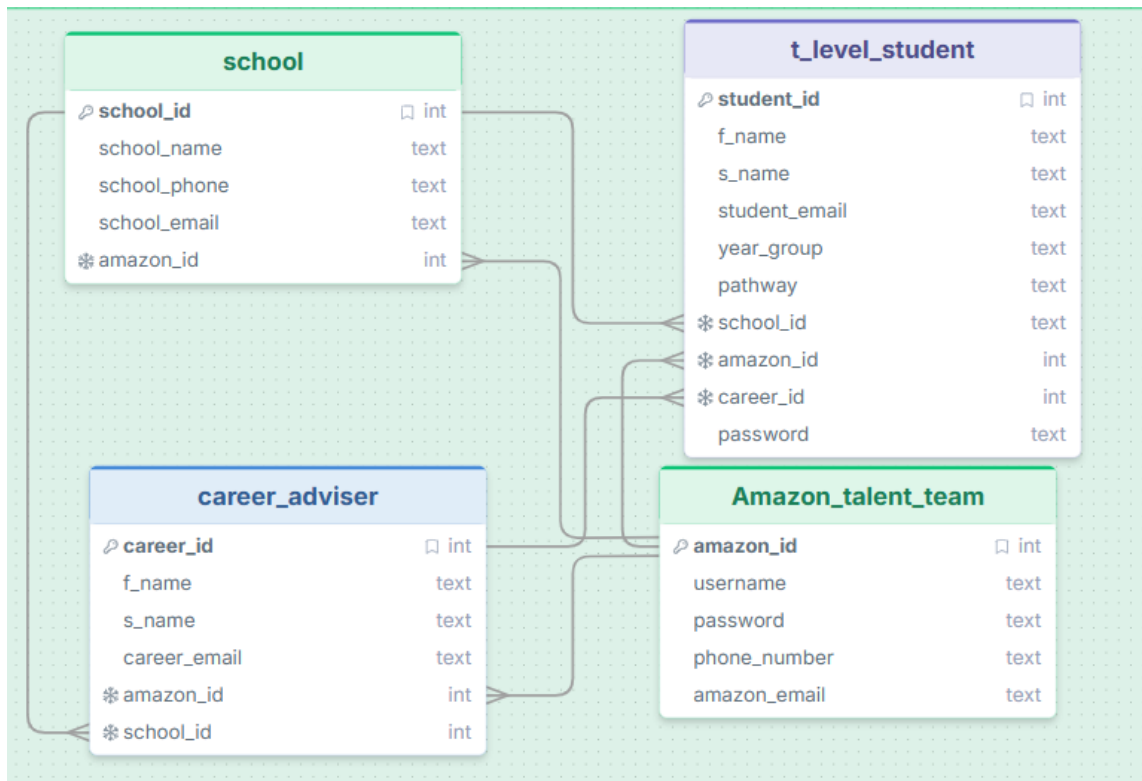
One to many relationship (ER)



Justification:

This ER diagram is a one-to-many relationship. This type of relationship is linked to one entity in a table is linked to multiple entities in another table. It is used to map out the entities that's related to each other within a database system and normally presented in a model with crow's foot notation. This may be the best way to design the database as it provides a visual presentation of the system as it lays out the information to the non-technical clients so they can easily understand how it works from the database design alone.

Database tables



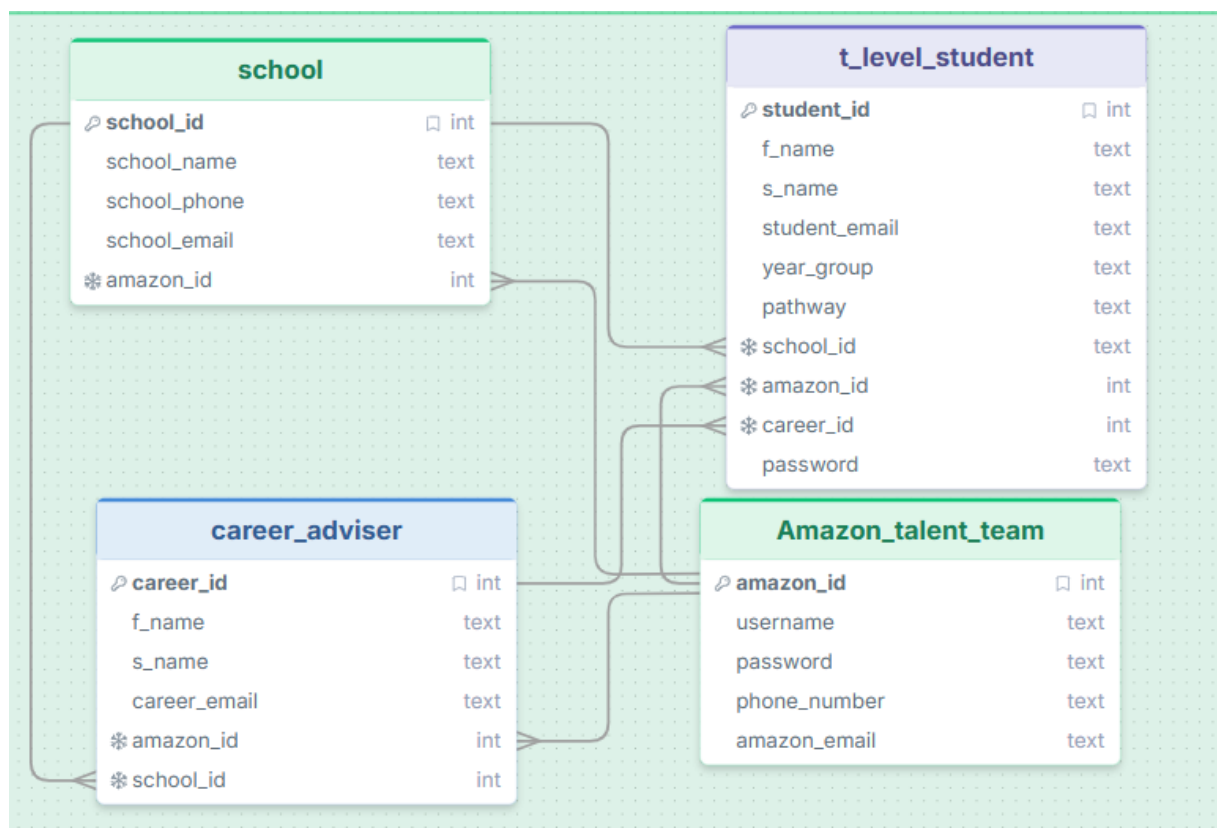
Justification:

- These are the database tables. The primary key is used as a unique identifier for each row of the database tables. This is used to ensure that there aren't any rows with the same key value (duplicated values) and no null values. When the student enters their login or register, the student ID are auto incremented so it can identify each student with a primary key.
- Foreign keys are used to link the two tables and maintain data consistency, ensuring that there aren't any invalid data entries to the database. It refers to the

primary key into another table so the foreign key could connect to the two tables to create a relationship to increase data integrity.

- These database tables are used to organize data to prevent data duplication as having duplicated information may cause data conflict and lead to errors when updating the table, improving data integrity. In addition, it stores attributes that specifically belong to the two entities rather than only one.
-

Data Dictionary



T_level_student

Field name	Data type	Description
student_id	Integer	Primary key of the field
f_name	varchar(255)	Name of the school

s_name	varchar(255)	School's phone number
student_email	varchar(255)	School's email
amazon_id	int	

School

Field name	Data type	Description
school_id	Integer	Primary key of the field
school_name	varchar(255)	Name of the school
school_phone	varchar(255)	School's phone number
school_email	varchar(255)	School's email
amazon_id	int	Foreign key of the field

Career_adviser

Field name	Data type	Description
career_id	Integer	Primary key of the field
f_name	varchar(255)	Name of the school
s_name	varchar(255)	School's phone number
career_email	varchar(255)	School's email
amazon_id	int	Foreign key of the field
school_id	int	Foreign key of the field

Amazon_talent_team

Field name	Data type	Description
Amazon_id	Integer	Primary key of the field
username	varchar(255)	Name of the school
password	varchar(255)	School's phone number
phone_number	varchar(255)	School's email
amazon_email	varchar(255)	Amazon's email

Justification:

- The data dictionary stores the information and relationships of data into a database table. Primary keys (IDs) are used as unique identifiers for each record in the database table, meaning they are distinguishing one record from another record in the same dataset. This ensures that the records don't share the same key and must be unique to prevent data conflict. Examples of a primary key is studentID- they're known to be the main table for the primary keys.
- Foreign keys (FK) are used to identify another row of another table. In other words, foreign keys are known as a second table as it handles a relationship between two tables. For example, schoolID is a foreign key in the t level student team. This is to connect between two entities and maintain to data integrity to invalid data entries to ensure data consistency.
- Varchar (variable character) is a variable character string in a fixated state. This is a flexible data type as anything can be entered in SQL and is widely used in database design. For example, phone numbers would contain various of string types such as letters and special characters.
- The data type integer is used to store whole numbers to represent numerical data without fractional parts which is commonly used in measurements and IDs. The data entered the field would be entered as integers in SQL databases and be able
- Boolean is a different type of data type that represents three logical values - TRUE, FALSE and UNKNOWN (known as NULL value). The NULL value is used to say that the column is unknown and not determined. If the inputs correlate to correction, the value would be true which applies the same principle to the FALSE value.

Data Flow Diagram

while in the index you can select the following:

1. Index: basic homepage where basic information is displayed allows you to access all over websites on the site.
2. About: gives information about the website and t-level offers and apprenticeships. Allowing users to gain more than the home: page information on what's at hand in the website and placements.

3. Contacts: displays all the ways to contact Amazon and us about the websites, help or support. as well as contacting us about more information.
4. Enroll: what allows the user to apply to t-level offers and apprentice ships. This gives a gateway for the user to apply for placements as well as gain more knowledge about how said placement will function, with additional information being location, date, length or place and level of apprenticeship or t-level offer.
5. Sign up: allows you to make an account and when signing up with a email and password. Users input their contact information into the database, so the user may be contacted when opportunities or issues arrive for or about placements.
6. T-level students: allows t-level students to view current apprenticeships and t-level offers that are more dedicated to there requirements.
7. Staff page: allows staff to view all users including students staff employment advisors and etc
8. settings: allows the user to tick boxes to help them meet said users disabilities needs

data flow information:

About:

the user can download content from the about section as well as gain information about t-level placements and offers

Contact:

Here the user can input there Email address and get additional information though email (using AWS services) to gain more information about t-level offers and apprenticeships

Enrolling:

1. A email is required for contacts so amazon can contact users about offers or over enquires
2. The users full name is also required so they maybe addressed properly when using the website or being emailed
3. Their year group is required so the system knows when they will be leaving t-level and when to remove them from the database
4. pathway to have more specialized advertisements

5. password for security so only the user with the password can log into said users account

Sign in:

1. User inputs there email and password
2. The database then compares passwords and to the one relevant to the email and checks if they are the same

T-level students:

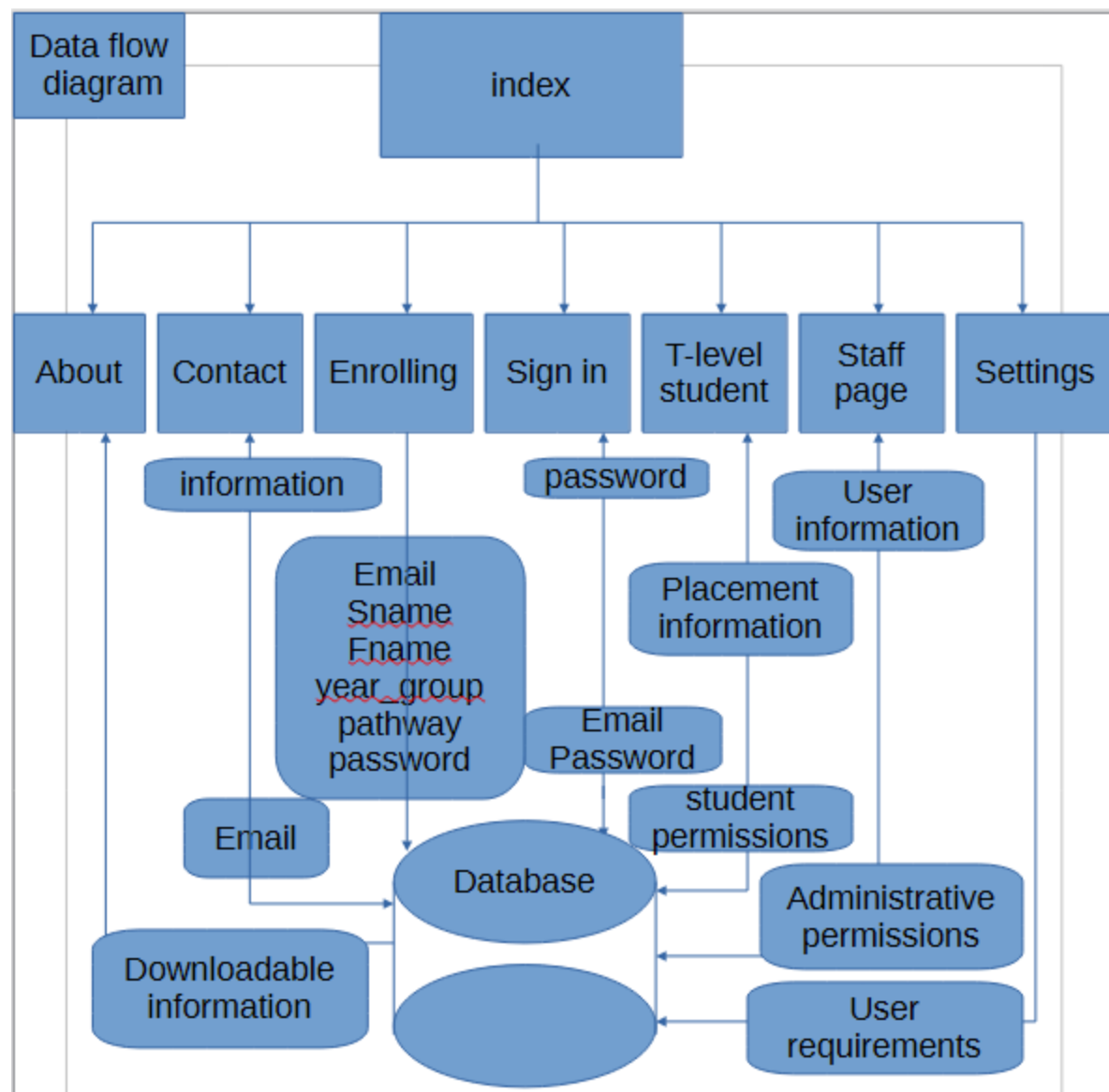
Firstly the Database checks if the user has student permissions. Then displaying dedicated information about t-level offers and apprenticeships if said user has student permissions else they can't access the information

Staff page:

Before the user can check the page they are checked for administrative access if they don't have admin they are kicked off and put back onto the index page. if said user has administrate access they are able to view all users in the database

Settings:

User can tick boxes based on said users requirements. They are saved onto the database depending on the user and then said requirements are met by changing the view of the website so that the website ca be used by everyone



Data Structures (Variable List)

Variable name	Scope	Data type	Where will it be used	Example data (PHP)	Reasoning

Dbcursor	Global (can be used as a parameter)	Database cursor type (object)	To the whole solution	N/A	There is a needed element to work with the database
\$servername	Local	string	Depends on where it has been declared from the solution but mainly to name the server	PHP - \$servername = "localhost"; // Global scope	To determine the name of the server host so the clients are on the right server host

\$username	local	string	When the users need to enter their username into the database server to be able to	<pre> PHP - \$username \$sql = "INSERT INTO amazon_talent_team (amazon_id,username,password,phone_number,amazon_email) VALUES (?, ?, ?,?, ?,?, ?,?)"; //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(2, \$_POST[username]); </pre>	So, the user's username would be entered and stored into the database
\$password	local	string	When the user tries to open the website this is hardcoded as 'Password' and is only to show that it was a customer rather than admin	<pre> PHP - \$password \$sql = "INSERT INTO amazon_talent_team (amazon_id,username,password,phone_number,amazon_email) VALUES (?, ?, ?,?, ?,?, ?,?)"; //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(3, \$_POST['password']); </pre>	The line of code that connects PHP to an SQL database take

			who accessed the database		s 4 arguments username and password are two of them and seem to be required to open said database
\$amazon_email	private	string	When the user is putting in their own email and when it gets recalled	PHP - \$amazon_email= echo \$_GET["amazon_email"]; \$password \$sql = "INSERT INTO amazon_talent_team (amazon_id,username,password,phone_number,amazon_email) VALUES (?, ?, ?, ?, ?, ?, ?, ?)"; //prepare the sql to be sent	The user's personal email would

			from the database	<pre>\$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(5, \$_POST[amazon_email]);</pre>	ld be entered and stored into the database
\$phone_number	private	string	When the user is putting in their phone number and when it gets recalled from the database	<pre>PHP - \$phone_number = echo \$_GET["phone_number"]; \$sql = "INSERT INTO amazon_talent_team (amazon_id,username,password,phone_number,amazon_email) VALUES (?, ?, ?, ?, ?, ?, ?, ?)"; //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(4, \$_POST[phone_number]);</pre>	This gets the Phone Number of the user from HTML which the n can be quality checked in

					PHP and the n added to the database
\$f_name	private	string	When the user is putting in their first name and when it gets recalled from the database	\$\$personal_email= echo \$_GET[f_name];	This gets the First Name of the user from HTML which the n can be quality checked in PHP and

					the n add ed to the dat aba se
\$school	priv ate	strin g	When the user is putting in their school and when it gets recalled from the database	PHP - \$school = echo \$_GET["school"];	This is req uire d to get the vari able of the user fro m the dat aba se tabl e
\$s_name	priv ate	strin g	When the user is putting in their surname and when it gets recalled	<pre> \$PHP = \$s_name= echo \$_GET["s_name"]; PHP - \$sql = "INSERT INTO t_level_student (school_id,f_name,s_name,email,password,ye ar,pathway,school_id,amazon_id,career_id) VALUES (?, ?, ?,?,?,?,?); //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql </pre>	This gets the surn ame of the user

			from the database	\$stmt->bindParam(3, \$_POST['s_name']);	from HTML which the n can be quality checked in PHP and the n added to the database
\$school_number	private	string	When the user is putting in their school's phone number and when it gets recalled	PHP - \$school_number = echo \$_GET["school_number"];	This gets the school phone number from

			from the database		m the user fro m HT ML whi ch the n can be qual ity che cke d in PHP and the n add ed to the dat aba se
\$year_group	private	string	When the user is putting in their year group and when it gets recalled	PHP - \$year_group= echo \$_GET["year_group"]; \$sql = "INSERT INTO t_level_student (school_id,f_name,s_name,email,password,year_group,pathway,school_id,amazon_id,career_id) VALUES (?, ?, ?,?,?,?,?); //prepare the sql to be sent	This gets the year group of the user

			from the database	<pre>\$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(6, \$_POST['year_group']);</pre>	from HTML which then can be quality checked in PHP and then added to the database
\$pathway	private	string	When the user is putting in their T-level pathway and when it gets recalled from the database	<pre>\$sql = "INSERT INTO t_level_student (school_id,f_name,s_name,email,password,year_group,pathway,school_id,amazon_id,career_id) VALUES (?, ?, ?, ?, ?, ?, ?, ?)"; //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(7, \$_POST[pathway]);</pre>	This gets the T-level course of the user

					from HTML which the n can be qual ity che cke d in PHP and the n add ed to the dat aba se
\$school_email	private	string	When the user is putting in their school email and when it gets recalled from the database	<pre> PHP - \$school_email = echo \$_GET["school_email"]; \$sql = "INSERT INTO school (school_id,school_name,school_phone,school_email, amazon_id) VALUES (?, ?,?, ?,?)"; //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(4, \$_POST[school_email]); </pre>	This gets the school email from the user

					from m HT ML whi ch the n can be qual ity che cke d in PHP and the n add ed to the dat aba se
<code>\$conn = my news qli</code>	N/A	string	Used to create a connectio n to MySQL	<code>\$conn = new mysqli(\$servername, \$username, \$password, \$dbname);</code>	To con nect to the serv er and be able to

					access data while connecting PHP to SQL
student_id	private	int(primary key)	Foreign key for the t level student table	<pre>\$sql = "INSERT INTO t_level_student (student_id,f_name,s_name,student_email,password,year_group,pathway,school_email,amazon_id,career_id,school_id) VALUES (?, ?, ?,?,?,,?)"; //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(1, \$_POST['student_id']);</pre>	To prevent school_id from being changed by binding the parameters for security.
career_id	private	int(primary key)	Foreign key for the	<pre>\$sql = "INSERT INTO career_adviser (career_id,f_name,s_name,career_email,,ama</pre>	To prev

		ry key)	career adviser table	<pre> zon_id,school_id) VALUES (?, ?, ?,?,?,?,?); //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(1, \$_POST[career_id]); </pre>	ent care er_i d fro m bein g cha nge d by bind ing the par ame ters for sec urity .
amaz on_id	priv ate	int(p rima ry key)	Auto- increment ed. The primary key for the database table(ama zon_talent _team)	<pre> \$sql = "INSERT INTO amazon_talent_team (amazon_id,username,password,phone_numbe r,amazon_email) VALUES (?, ?, ?,?,?,?,?,?)"; //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(1, \$_POST[amazon_id]); </pre>	To prev ent ama zon _id fro m bein g cha nge d by bind ing the

					parameters for security .
school_id	private	int(primary key)	Auto-incremented. The primary key for the database table(client)	<pre> \$sql = "INSERT INTO school (school_id,school_name,school_phone,school_email, amazon_id) VALUES (?, ?,?, ?,?)"; //prepare the sql to be sent \$stmt = \$conn->prepare(\$sql); //prepare to sql \$stmt->bindParam(1, \$_POST[school_id]); </pre>	To prevent school_id from being changed by binding the parameters for security .

Algorithm Designs (Pseudo code)

connecting PHP to SQL

```
$servername = "localhost"; //sets servername
```

```
$dbusername = "root"; // had to change this variable name as it fought against the  
admin reg and user reg
```

```
$dbpassword = ""; //password for database useraccount
```

```
$dbname = "digital_solution"; //database name to connect to
```

```
try { //attempt this block of code, catching an error
```

```
    $conn = new PDO("mysql:host=$servername;port=3306;dbname=$dbname",  
$dbusername, $dbpassword); // creates a PDO connection to the database
```

```
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION); //sets  
error modes
```

```
    return $conn;
```

```
} catch (PDOException $e) { //catch statement if it fails
```

```
    error_log("Database error in super_checker: " . $e->getMessage());
```

```
    // Throw the exception
```

```
    throw $e; // Re-throw the exception // outputs the error
```

//links PHP to database, enables editing of already existent table and shows if table
connection has worked

error messages

```
} catch(PDOException $e){
```

```
    echo $e->getMessage();
```

```
    exit;
```

```
} catch(Exception $e) {
```

```
    echo $e->getMessage();
```

```
    exit;
```

```
}
```

//returns an error message when code crashes instead of just closing window

Registration

```
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    if ($_POST["password"] == $_POST["c_password"]) {
        if (onlyuser(dbconnect_insert(), $_POST["email"])) {
            if (reg_user(dbconnect_insert())) {
                $_SESSION["usermessage"] = "You are successfully registered.";
                header("Location: login.php");
                exit;
            } else {
                $_SESSION["usermessage"] = "There was an error registering your account.";
                header("Location: register.php");
                exit;
            }
        } else {
            $_SESSION["usermessage"] = "Email not unique.";
            header("Location: register.php");
        }
    } else {
        $_SESSION["usermessage"] = "Passwords do not match.";
        header("Location: register.php");
        exit;
    }
}
```

login page login check

```
try {
    if ($_SERVER["REQUEST_METHOD"] == "POST") {
        if (!isset($_SESSION["student_id"])) {
            $_SESSION["usermessage"] = "You are already logged in.";
            header("Location: login.php");
            exit;
        } elseif ($_SERVER["REQUEST_METHOD"] == "POST") {
            $usr = login(dbconnect_insert(), $_POST["email"]);
```

```

if($usr && password_verify($_POST["password"], $usr["password"])) {
    $_SESSION["usermessage"] = "You are now logged in.";
    $_SESSION["student_id"] = getUserid(dbconnect_insert(), $_POST["email"]);
    header("Location: index.php");
    exit;
} elseif (!$usr) {
    $_SESSION["usermessage"] = "invalid email or password.";
    header("Location: login.php");
} else{
    $_SESSION["usermessage"] = "something went wrong.";
    header("Location: login.php");
    exit;
}
}
}
}

```

school getter

```

try {
    $schools = school_getter(dbconnect_insert());
    echo '<select name="school_id" id="school_id">' . "\n";

    echo '  <option value="">-- Please select a school --</option>' . "\n";

    foreach ($schools as $school) {

        echo '  <option value="' . $school["school_id"] . '">' . $school["school_name"] .
        '</option>' . "\n";
    }

    echo '</select>';
}

```

```

    } catch (PDOException $e) {
        echo $e->getMessage();
        exit;
    } catch (Exception $e) {
        echo $e->getMessage();
        exit;
    }
}

```

Test Strategy

Description of test	Test data to be used	Expected outcome	Actual outcome	comments
Describe why test is being conducted	What data is being entered to test it	Intention of result	What happened? Include image	How to mitigate?

Here is the proposed test log we are going to use throughout the task when testing. It follows the colour scheme of amazon and includes key features of a test log. The data that needs to be entered has been described to remove confusion.

Additionally, the headings are what is needed for a complete test log. Description of the test is needed, so the reader knows why this test has been done. Test data needs a varied amount so we can see valid, extreme, and erroneous results. We need to know what the tester was expecting to happen next so if it's not the actual outcome, it can start to be mitigated. The actual outcome needs a brief explanation of what happened and an image to show that too. Comments show intended actions to resolve a potential problem. Only needed if the test fails. We plan to test, create a test log, and start testing our solution in the third week of our project to give us time to go back and fix any errors to make our solution more robust.

[test log template.docx](#)

[test log template.docx](#)